📘 MODULE 23: PROPS.CONF + TRANSFORMS.CONF

23.1 Concept

- **`props.conf`** attaches settings to a **sourcetype, source, or host** (`[sourcetype]`, `[source::/path/*]`, `[host::name]`).
- **`transforms.conf`** defines **reusable regex-based operations**. `props.conf` **calls** them by name.

```
props.conf:       [my_app_logs]  TRANSFORMS-drop = drop_debug
                                                   │
transforms.conf:  [drop_debug]   REGEX = ...   ◄┘
```

23.2 Index-time vs search-time (the key to this module)

Pipeline stage	Examples	Where it runs
Index-time (parsing)	Line breaking, timestamps, routing, filtering, masking, sourcetype/host/index override	First full instance (HF or indexer)
Search-time	`EXTRACT-`, `REPORT-`, `FIELDALIAS-`, `EVAL-`, `LOOKUP-`	Search head

Index-time changes only affect **new** events. Search-time changes affect old events too.

23.3 Core `props.conf` settings

Event breaking and timestamps (the "must-have" set for a custom sourcetype):

ini

```ini
[my_app_logs]
SHOULD_LINEMERGE = false
LINE_BREAKER = ([\r\n]+)
TIME_PREFIX = ^
TIME_FORMAT = %Y-%m-%d %H:%M:%S
MAX_TIMESTAMP_LOOKAHEAD = 19
TZ = UTC
TRUNCATE = 10000
```

Setting	Meaning
`SHOULD_LINEMERGE = false`	Don't merge lines; use `LINE_BREAKER`
`LINE_BREAKER`	Regex for where events split. The **capture group** is the discarded delimiter.
`TIME_PREFIX`	Regex marking where the timestamp **begins**
`TIME_FORMAT`	`strptime` pattern of the timestamp
`MAX_TIMESTAMP_LOOKAHEAD`	How many characters after the prefix to read for the timestamp (here 19 = length of `2026-11-08 10:15:01`)
`TZ`	Time zone of the log, if it has none
`TRUNCATE`	Max bytes per event (default 10000; 0 = unlimited)

Search-time settings:

ini

```ini
[my_app_logs]
KV_MODE = auto
EXTRACT-ip = ip=(?<src_ip>\d{1,3}(?:\.\d{1,3}){3})
REPORT-user = extract_user
FIELDALIAS-src = src_ip AS src
EVAL-action = if(status=="success","success","failure")
LOOKUP-users = users_lookup username AS user OUTPUT department
```

- `EXTRACT-<name>` is an inline regex (named capture groups).
- `REPORT-<name>` points to a `transforms.conf` stanza.
- `FIELDALIAS-`, `EVAL-` (calculated field), `LOOKUP-` (automatic lookup) feed CIM (Module 17).

Index-time operations:

ini

```ini
[my_app_logs]
TRANSFORMS-drop = drop_debug
SEDCMD-mask = s/password=\S+/password=****/g
```

`SEDCMD-` does inline sed-style masking at index time, with no transform needed.

23.4 `transforms.conf` attributes

Attribute	Meaning
`REGEX`	The pattern to match (capture groups available as `$1`, `$2`)
`SOURCE_KEY`	**Which field to apply the regex to** (default `_raw`)
`DEST_KEY`	**Where to write the result** (e.g. `queue`, `_MetaData:Index`, `MetaData:Sourcetype`, `MetaData:Host`, `_raw`)
`FORMAT`	**What to write** (e.g. `nullQueue`, `indexQueue`, `sourcetype::x`, `host::$1`, `field::$1`)
`WRITE_META`	Make the extraction an **indexed field** (index-time)
`DEFAULT_VALUE`	Used if the regex doesn't match

Mental model: **"Look in SOURCE_KEY, match REGEX, then write FORMAT into DEST_KEY."**

23.5 Realistic worked examples

Sample log (`/var/log/myapp/app.log`):
`2026-11-08 10:15:01 INFO user=alice action=login status=success ip=10.0.0.5 password=Secret123`

Example 1: override sourcetype by path

ini

```ini
# props.conf
[source::/var/log/myapp/*.log]
TRANSFORMS-st = set_myapp_st

# transforms.conf
[set_myapp_st]
REGEX = .
DEST_KEY = MetaData:Sourcetype
FORMAT = sourcetype::my_app_logs
```

What it does: every event from that path gets the sourcetype `my_app_logs`.
Mistake: forgetting the `sourcetype::` prefix.

Example 2: drop events (nullQueue)

ini

```
# props.conf
[my_app_logs]
TRANSFORMS-drop = drop_debug

# transforms.conf
[drop_debug]
REGEX = \sDEBUG\s
DEST_KEY = queue
FORMAT = nullQueue
```

Dropped events do **not** count against your license (they're discarded before indexing).

Example 3: keep only certain events (drop all, then keep)

ini

```
# props.conf
[my_app_logs]
TRANSFORMS-filter = setnull, keep_errors

# transforms.conf
[setnull]
REGEX = .
DEST_KEY = queue
FORMAT = nullQueue

[keep_errors]
REGEX = \s(ERROR|WARN)\s
DEST_KEY = queue
FORMAT = indexQueue
```

Order matters: transforms run left to right, so `setnull` drops everything, and then `keep_errors` re-queues the matches.

Example 4: mask a password

ini

```
[my_app_logs]
SEDCMD-mask = s/password=\S+/password=****/g
```

Why: compliance. Original values never reach the index. **Mistake:** testing on old events, because only new data is masked.

Example 5: route to another index by content

ini

```
# props.conf
[my_app_logs]
TRANSFORMS-route = route_failed

# transforms.conf
[route_failed]
REGEX = status=failure
DEST_KEY = _MetaData:Index
FORMAT = security
```

The `security` index must exist.

Example 6: set the host from the event text

```ini
[set_host_from_event]
REGEX = host=(\S+)
DEST_KEY = MetaData:Host
FORMAT = host::$1
```

`$1` is the first capture group.

Example 7: search-time extraction via transforms

```ini
# props.conf
[my_app_logs]
REPORT-user_action = extract_user_action

# transforms.conf
[extract_user_action]
REGEX = user=(\w+)\s+action=(\w+)
FORMAT = user::$1 action::$2
```

23.6 Where do these files go?

Setting type	Put it on
Parsing, filtering, routing, `SEDCMD`	**Indexer or HF** (the first full instance), not the UF
`EXTRACT-`, `REPORT-`, `FIELDALIAS-`, `EVAL-`, `LOOKUP-`, `KV_MODE`	**Search head** (and sometimes indexers in distributed setups)
`INDEXED_EXTRACTIONS` (structured data like CSV/JSON)	The **UF** (the exception)

23.7 Test and verify

```bash
sudo /opt/splunk/bin/splunk btool props list my_app_logs --debug
sudo /opt/splunk/bin/splunk btool transforms list drop_debug --debug
sudo /opt/splunk/bin/splunk btool check
```

Then restart (index-time) and feed **new** events. Check:

```spl
index=main sourcetype=my_app_logs | eval lag=_indextime-_time | stats count avg(lag)
```

Also: Settings → **Add Data** → **Upload** shows a preview of how `props.conf` settings break events and parse timestamps without indexing real data. This is the safest way to test.

23.8 Hands-on lab

1. Create the sample file:

```bash
mkdir -p /tmp/myapp
cat >> /tmp/myapp/app.log << 'EOF'
2026-11-08 10:15:01 INFO user=alice action=login status=success ip=10.0.0.5 password=Secre
2026-11-08 10:15:02 DEBUG heartbeat ok
2026-11-08 10:15:03 ERROR user=bob action=login status=failure ip=10.0.0.9 password=Hunter
EOF
chmod -R a+rX /tmp/myapp
```

2. `etc/system/local/inputs.conf`: `[monitor:///tmp/myapp] index=main sourcetype=my_app_logs`
3. Add the `my_app_logs` props, the `drop_debug` transform, and `SEDCMD-mask` from above.
4. Restart. Append two more lines and search `index=main sourcetype=my_app_logs`.
5. **Expected:** no DEBUG lines, passwords show `****`, timestamps match the log, and `user`, `action` fields exist.

23.9 Troubleshooting: "props/transforms not working"

Symptom	Cause
Nothing changes	Wrong tier (placed on UF or the wrong server), no restart
Works in btool, not in data	Testing on old events
Transform never runs	Name in `TRANSFORMS-x = name` doesn't match `[name]` in `transforms.conf`
Stanza doesn't match	Props stanza is for a sourcetype/source that differs from what events actually have
Regex doesn't match	Test it in a search with `rex` first (adapt syntax)
Wrong timestamp	`TIME_PREFIX` / `TIME_FORMAT` / `MAX_TIMESTAMP_LOOKAHEAD` mismatch
Fields missing after change	Search-time config not shared, or on the wrong app

23.10 Interview Q&A

- **props vs transforms?** props attaches settings to a sourcetype/source/host; transforms holds reusable regex rules that props calls.
- **How do you mask data before indexing?** `SEDCMD-` in props, or a transform rewriting `_raw`, on the first full instance.
- **How do you drop events?** A transform with `DEST_KEY = queue` and `FORMAT = nullQueue`.
- **What are SOURCE_KEY and DEST_KEY?** Where to read from and where to write to.
- **Why doesn't my change apply to old data?** Index-time changes only affect new events.
- **Index-time vs search-time extraction?** Search-time is flexible and preferred; index-time is fixed and costs storage.
- **Props on UF?** Generally ineffective for parsing, except `INDEXED_EXTRACTIONS`.

23.11 Revision

props attaches, transforms defines · index-time on HF/indexer, search-time on SH · `nullQueue` drops, `indexQueue` keeps · `SEDCMD` masks · transform order matters · test with new events, verify with btool.